Speaker Information

[Name] Nao Nakagawa-Tamagawa 玉川（中川） 直

[Institute] Kagoshima University 鹿児島大学

[Position] Assistant Professor

[Email Address] naonaka@m.kufm.kagoshima-u.ac.jp

[Lab / Personal Website]

https://researchmap.jp/chockie/?lang=en

Talk Information

My Specialized Techniques as a Seed of Synergy
- **In Utero Electroporation (IUE)** & **Patch-Clamp Recording**

[Abstract] I will introduce my experimental techniques and some past/current works to seek for seeds of collaboration. **IUE** can directly introduce multiple plasmids into a specific subset of neurons in the mouse neocortex. (1) Various mutants can be co-expressed with fluorescent proteins, enabling the efficient screening of the effects of mutations. No transgenic mice needed; only plasmids. (2) Using CRISPR-Cas9 technology, genes of interest can be knocked out in a specific subset of neurons. Genes that would be lethal with whole-body knockout are also applicable. **Patch-Clamp Recording** is my expertise, with >15 years of experience and five collaborative publications. I can record neuronal membrane properties, firing pattern, synaptic connection, gap junction, and evoked and spontaneous synaptic inputs from acute brain slices and dissociated primary culture neurons. I can also record channel currents from cell lines.

Research Overview (for collaboration discussion)

[Research Interests] Neuronal morphogenesis and circuit formation in neocortex

[Current Research Projects] (1) Analyzing how epilepsy-related Nav1.2 mutations affect channel properties, neuronal morphology, migration, and axonal projection in mouse neocortex. (2) Mechanism of soma-to-AIS structure determination and remodeling.

[Techniques] in utero electroporation, patch-clamp recording, dissociated neuron primary culture

[Aspects or techniques of interest for future collaboration] I'm open to any form of international joint research and collaboration. I want to discuss with researchers on topics related to mine or who have techniques applicable to my research (calcium signaling analysis, time-lapse confocal imaging of dissociated culture neurons, neuron or circuit modelling, etc).

Short Bio

• 2006 – 2011. Ph.D. student, Department of Physiological Sciences (belonging to NIPS), School of Life Science, SOKENDAI
• 2011 – 2018. Researcher, Laboratory for Local Neuronal Circuits, RIKEN BSI
• 2018 – Now. Assistant professor, Grad Sch Med & Dent Sci, Kagoshima Univ.